

## Digital Logic Design

### List of questions

The chapters are mentioned according to the book "Digital System Principal & Application" by Ronald J Tocci & Neal S Widmer

## INTRODUCTORY CONCEPTS

### CHAPTER-1

- (1) What is Analogue System & Digital system? What are the difference among Analogue & Digital system?
- (2) What are the advantages of Digital system over Analogue system? What are the imitations of Digital system?
- (3) What are the Digital number systems? Describe them briefly.

## NUMBERS SYSTEMS & CODES

### CHAPERT-2

- (1) What are the Digital number systems? Describe them briefly.
- (2) What is BCD code? Distinguish between BCD and Binary representation.
- (3) Discuss how Parity method is used for error detection? Describe a 4-bit parity generator circuit with diagram.
- (4) \*\*\*The Number Conversion Problems of this chapter are very important.\*\*\*
- (5) What is ANSI code?
- (6) What are Byte, Nibble and Word?

## DESCRIBENG LOGIC CIRCUITS

### CHAPTER-3

- (1) What are the basic Boolean operation? Explain them with truth tables. Give the symbolic representations of the Gates.
- (2) \*\*\*Proofs of logic expressions using Boolean theorems are very important\*\*\*
- (3) Which gates are known as universal gates? Describe them.
- (4) Sate and prove Demorgan"s theorem.
- (5) Construct INVERTER, AND gate and OR gate using NAND gate and Nor gate.
- (6) Discuss the alternate representation of logic gates.

## COMBINATIONAL LOGIC CIRCUITS

### CHAPTER-4

- (1) Discuss the two forms representation of logic circuit.
- (2) \*\*\*The problems of simplifying logic expressions and circuits are important.\*\*\*
- (3) Define Combinational logic circuit, Sequential logic circuit, Karnaugh Map Method, Looping groups of Pairs, Looping groups of Quads and looping groups of Octets.

- (4) Describe exclusive OR and exclusive NOR circuits with truth table and circuit diagram.
- (5) Describe parity Generator and checker circuit with example and circuit diagram.
- (6) 4-bit\5-bit\6-bit parity generator circuit and checker diagram.
- (7) \*\*\*Problems of Karnaugh graph are very important.\*\*\*

### FLIP-FLOPS AND RELATED DEVICES

#### CHAPTER-5

- (1) Describe NAND latch and NOR latch with truth table and circuit diagram.
- (2) Define: Asynchronous System, Synchronous System, Clock signal, Positive going transition, Negative going transition, Setup time and Hold time.
- (3) Describe Clocked S-R flip flop with internal diagram.
- (4) Describe Clocked J-K flip flop with internal diagram.
- (5) Describe Clocked D flip flop with internal diagram.
- (6) Discuss Propagation Delays, Maximum Clocking frequency and Low times in term of Flip-flops.
- (7) Mention the difference between Asynchronous and synchronous input.
- (8) Mention the applications of Flips-flops,
- (9) What is the race around the problem in Flip-flops.

### DIGITAL ARITHMATICS: OPERATIONS AND CIRCUITS

#### CHAPTER-6

- (1) Write short notes on the following with example: Binary addition, Sign magnitude system, 2's complement system, BSD addition & subtraction.
- (2) \*\*Problems of addition & subtraction in 2's complement system are very important.\*\*
- (3) What is parallel binary adder circuit? Draw a 5-bit Parallel binary adder circuit.
- (4) Describe a Full adder circuit with internal diagram.
- (5) Describe a half adder circuit with internal diagram.
- (6) Describe 4-bit parallel adder circuit.
- (7) Describe 2's complement Addition & Subtraction circuit.
- (8) Describe ALU with diagram.

### COUNTERS & RESISTOR

#### CHAPTER-7

- (1) Describe Asynchronous (Ripple) counter. Discuss Propagation delay(Timing problem) of Asynchronous (Ripple) counter.
- (2) Describe Synchronous (Parallel) Counter with diagram.
- (3) Draw the diagram of a Synchronous MOD-6(or 16 or 13 etc.) counter.
- (4) Describe Synchronous Up/Down counter with necessary diagram.
- (5) Describe Presetable Synchronous counter with necessary diagram.
- (6) Describe Ring counter with necessary diagram.

- (7) Describe Johnson counter with necessary diagram.
- (8) Describe BCD counter with necessary diagram.
- (9) How would you connect two 74ALS174s to operate as 12-bit shift register?

## MISLOGIC CIRCUIT

### CHAPTER-8

- (1) What is Decoder? Describe 3 lines to 8 line decoder with necessary diagram.
- (2) What is Multiplexer? Describe 8 input multiplexer with diagram & truth table.
- (3) What is De-multiplexer? Describe 8 input multiplexer with diagram & truth table.
- (4) What is encoder? Describe 8 line to 3 line Encoder with diagram & truth table.
- (5) Mention the uses of Multiplexer.
- (6) Discuss the use of Multiplexer in Parallel to Series conversion.
- (7) Discuss the use of Multiplexer in Implementation of logic expression.
- (8) Mention the application of Encoder and Decoder.
- (9) Mention the application of Multiplexer and De-multiplexer.

## INTERFACING WITH THE ANALOGUE WORD

### CHAPTER-9

- (1) Describe the operation of Digital to Analogue converter.
- (2) Describe D'A converter circuitry.
- (3) Describe the operation of Analogue to Digital converter.
- (4) Describe A/D converter circuitry.